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TrustCheck PK: A Pre-Purchase Trust Checker for Social Commerce in Pakistan

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APPLICATION: TRUSTCHECK PK
ASSESSMENT: ASSIGNMENT 02 – MVP DEVELOPMENT

1. Introduction

TrustCheck PK is a mobile-first website application created to help consumers evaluate a new online vendor before purchasing from that merchant. This application was based on the idea and design in Figma as described in Assessment 1. The major purpose of this application is to give consumers an easy to follow check journey, a trust snapshot, evidence of the assessment and safer purchasing advice. MVP was concentrated on showcasing the core experience of the application rather than its ability to store data permanently and detect fraudulent activities (Han et al., 2018). Cline was used in Visual Studio Code to examine, analyze and develop the current application.

2. MVP Overview

This is illustrated through the use of the TrustCheck PK MVP whereby the buyer goes through the process of making an informed purchase before buying from the unknown seller via social commerce and independent online sellers. The first step here is to log into the application and provide details about the unknown seller (Lima & da Silva, 2024).

The main application flow is:

Login → Seller Check → Seller Details → Trust Snapshot → Evidence and Checklist → Decision Summary

Information about the users and sellers is not stored in the MVP. It helps to keep the prototype concentrated on proving the key functionality and user experience.

The application is designed around the central product message:

“Check the seller before you pay.”

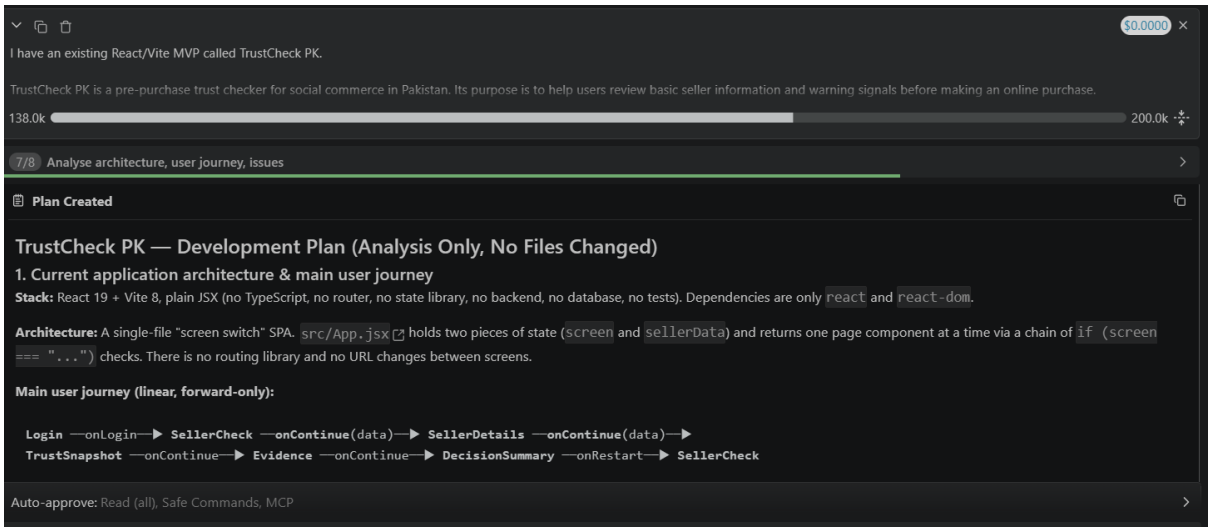
3. Cline Development Process

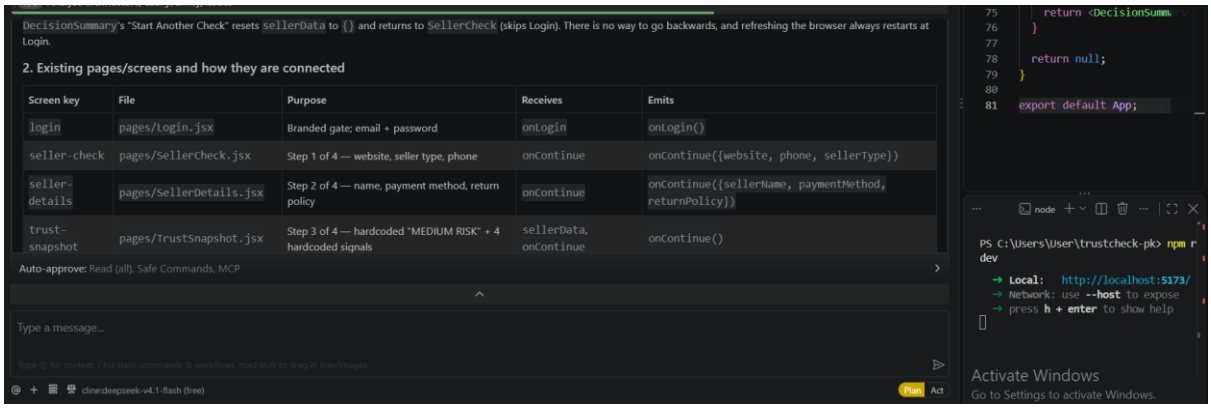
3.1 Cline Plan Mode

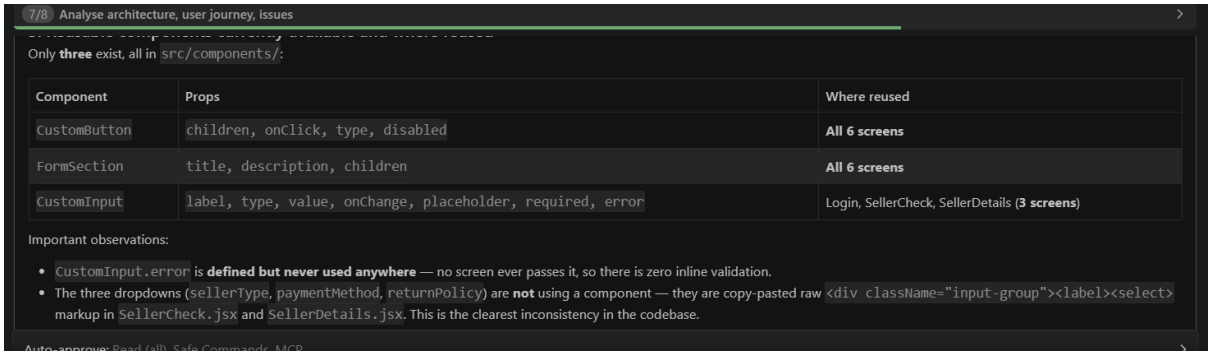
The Cline Plan Mode was implemented prior to implementing any major changes to the application. Since the baseline of the TrustCheck PK application was already functioning, the first task was to evaluate the current state of affairs without changing the functioning code.

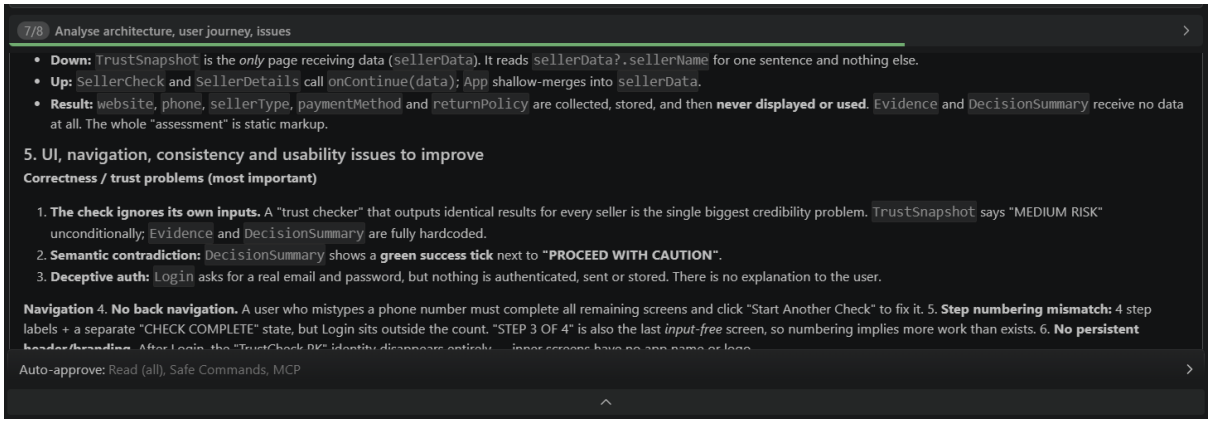
The first step in the planning process included instructing Cline to analyze the current state of affairs including the architecture of the application, its pages, re-usable components, styles and data flow. Cline had to evaluate how it is possible to implement the assessment criteria for the re-usable components and 21st.dev.

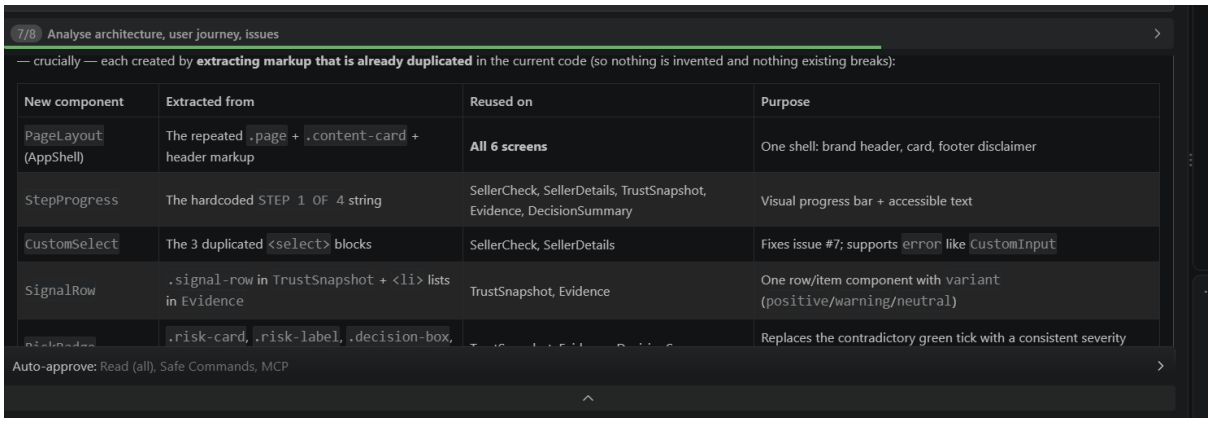
Plan Mode proved to be effective because it allowed separating the planning and implementation processes. Rather than implementing changes immediately, one could evaluate the current situation first and make a development plan.







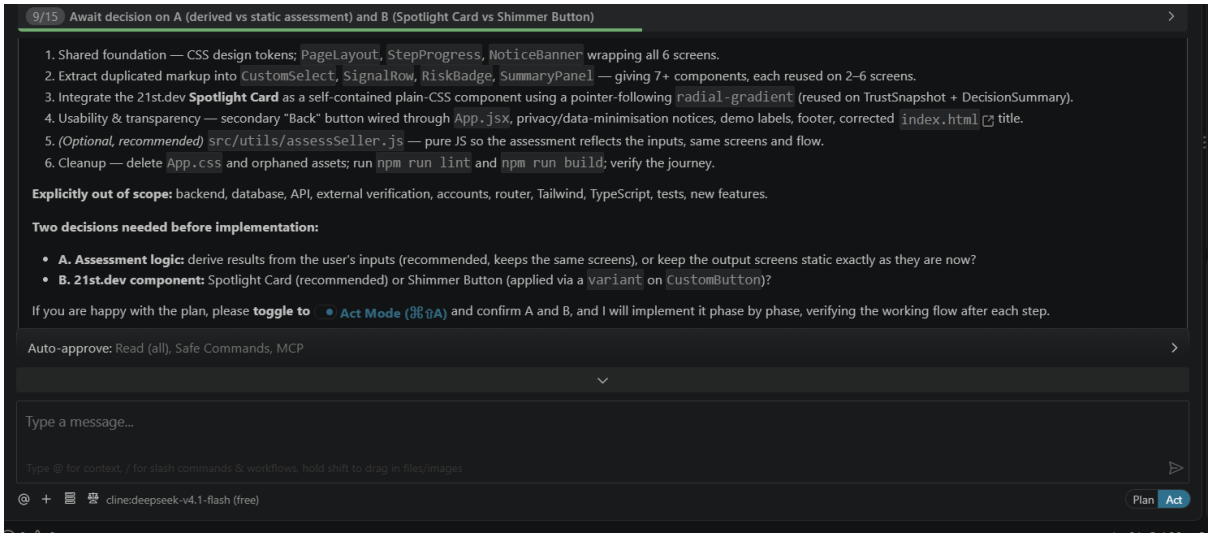




3.2 Cline Act Mode

Following the development of the plan, the implementation of the changes was done using Cline Act Mode. The Act Mode was used to make modifications to the React project, improve the reusability of the components, keep the consistency between screens, and incorporate the required third-party UI component.

The method used to implement the changes was through iteration as opposed to asking Cline to redo the entire program from scratch. This ensured that the existing functionality of TrustCheck PK is maintained as its architecture and user interface were improved. Act Mode was used to debug and refine the application as necessary. After making any modifications, the application was tested on the web browser.



4. Starting Prompt

4.1 Full Starting Prompt

The following prompt was used to begin the Cline planning process:

I have an existing React/Vite MVP called TrustCheck PK.

Introduction TrustCheck PK is a pre-purchase trust checker designed for social commerce in Pakistan. The goal of the application is to enable buyers to evaluate the basic information about sellers and any warnings.

The application is already working, so do not modify any files yet.

First, inspect the existing project structure and analyse the current application.

Please provide a development plan covering:

1. The current application architecture and main user journey.
2. The existing pages/screens and how they are connected.
3. The reusable components currently available and where they are reused.
4. How data is passed between App.jsx and the page components.
5. Any UI, navigation, consistency or usability issues that should be improved.

6. How to strengthen the application so it clearly demonstrates at least three custom reusable UI components reused across at least two screens each.

7. How one additional UI component from 21st.dev could be integrated into the existing application without unnecessarily changing the working functionality.

8. Improvements needed to make the MVP look professional and consistent with the TrustCheck PK purpose.

9. Any potential privacy, transparency or data-minimisation considerations that should be reflected in the interface.

Important requirements:

- The existing working functionality should be preserved.
- Do not introduce a backend or database.
- Do not add unnecessary features outside the MVP scope.
- Do not modify files during this planning stage.
- Explain your proposed changes before implementation.
- Keep the solution suitable for a beginner React/Vite project.

Focus only on analysis and planning at this stage.

4.2 Starting Prompt Annotation

The initial prompt had been framed in a way that gave Cline all the information about the purpose of the app and its technical requirements for assessment. First, the prompt made Cline aware of what TrustCheck PK does so that he gets an idea of how the user flow is going to be before making any changes. Furthermore, the prompt did not allow Cline to change the files as the application was fully working and only the development plan had to be framed based on the requirements.

5. Follow-Up Prompts

The development process used follow-up prompts to continue the inspection and guide implementation.

5.1 Follow-Up Prompt 1 – Complete Project Inspection

Prompt used:

Continue the inspection. Please read all files in src/pages, all files in src/components, src/index.css, and package.json. Do not modify any files. After reading them, provide the complete architecture analysis and development plan.

Problem addressed:

This prompt ensured that Cline did not base its recommendations only on App.jsx. It instructed Cline to inspect the page components, reusable components, styling and project configuration before producing the development plan.

5.2 Follow-Up Prompt 2 – Development Plan

Prompt used:

Based on the files you have inspected, now provide the complete development plan.

Do not modify any files.

Your plan must specifically address:

1. The TrustCheck PK user journey.
2. The three reusable components CustomButton, CustomInput and FormSection.
3. How each component will be reused across at least two different screens.
4. The UI/UX improvements required for the MVP.
5. Integration of one 21st.dev component.
6. Privacy, consent, transparency and data-minimisation considerations.
7. How the existing working functionality will be preserved.

8. The specific files that would need to be changed during implementation.

Present the plan clearly in numbered steps.

Problem addressed:

This prompt focused Cline's final planning response on the specific assessment requirements. It also made component reuse, third-party integration, ethical considerations and preservation of existing functionality explicit development goals.

5.3 Follow-Up Prompt 3 – Implementation Refinement

Problem addressed:

This prompt was used to implement and refine the planned changes in the working application. The final version of this section should contain the exact prompt from the Cline conversation used for the implementation stage.

6. Custom Reusable UI Components

The assessment requires at least three custom reusable UI components. Each component must be used on at least two different screens. TrustCheck PK uses reusable components to maintain consistency and reduce repeated interface code.

6.1 CustomButton

Component: CustomButton

Purpose:

The CustomButton component provides a consistent button style and interaction pattern throughout the application. Instead of creating separate button designs for every screen, the same component can be reused with different text, actions and properties.

Example reuse:

- Login screen
- Seller Check screen

- Seller Details screen
- Evidence screen
- Decision Summary screen

Prompt used to create the component:

Create a reusable React CustomButton component for the TrustCheck PK application. The component should accept props for button text, click behaviour, disabled state and appropriate styling variants. Keep the component simple and reusable so it can be used consistently across multiple screens. Do not duplicate button implementation in individual pages.

Assessment evidence:

The same CustomButton component is visibly used on multiple screens in the browser.

6.2 CustomInput

Component: CustomInput

Purpose:

The CustomInput component provides a consistent input field for collecting seller-related information. It can accept props such as label, placeholder, value, change handler and input type.

Example reuse:

- Login screen
- Seller Check screen
- Seller Details screen

Prompt used to create the component:

Create a reusable React CustomInput component for TrustCheck PK. It should accept props for label, placeholder, value, onChange, type and error or helper text where appropriate. The

component should have consistent styling and accessibility-friendly labelling. Make it reusable across the login and seller information screens.

Assessment evidence:

The same input component is reused for different types of information across multiple screens.

6.3 FormSection

Component: FormSection

Purpose:

The FormSection component provides a consistent structure for grouping related form information. It helps maintain spacing, headings and visual organisation across different screens.

Example reuse:

- Seller Check screen
- Seller Details screen
- Evidence screen
- Other information-entry screens where appropriate

Prompt used to create the component:

Create a reusable React FormSection component for TrustCheck PK that groups related form content into a consistent visual section. The component should accept children and suitable props such as title or description so that different pages can reuse the same structure without duplicating layout code.

Assessment evidence:

The same FormSection component appears on multiple screens with different content passed through its props.

7. 21st.dev Component Integration

A UI component from 21st.dev was integrated into the TrustCheck PK application as an additional third-party interface component.

The component was selected because it supports the visual presentation of the TrustCheck PK interface without changing the core purpose of the application. The component was adapted to fit the existing application styling and user journey.

8. Component Reusability Demonstration

The following table summarises the reuse of the three custom components.

Custom Component	**Screen 1**	**Screen 2**	**Other Reuse**
CustomButton	Login	Seller Check	Seller Details, Evidence
CustomInput	Login	Seller Check	Seller Details
FormSection	Seller Check	Seller Details	Evidence

The reuse of these components demonstrates a UI-kit approach. Instead of creating separate versions of similar interface elements for every page, common elements are defined once and reused through React components and props.

9. React Props and Data Flow

React props allow a parent component to pass information or functions to a child component. In TrustCheck PK, this approach is used to allow reusable components and screens to receive different data without duplicating their implementation.

For example, a parent component can pass seller information and an event handler to a child Trust Snapshot component:

```
<TrustSnapshot
  sellerData={sellerData}
  onContinue={handleEvidence}
```

/>

The child component can receive these values through its props:

function TrustSnapshot({ sellerData, onContinue }) {

In this example, sellerData contains information required by the Trust Snapshot screen, while onContinue provides the function that moves the user to the next stage of the journey. The parent controls the data and behaviour, while the child component is responsible for presenting the information and responding to the supplied function.

10. Ethical Considerations

Ethical considerations become crucial when developing TrustCheck PK since the project deals with information connected to online sellers and may involve some sensitive purchase decisions (Olivia, 2018).

10.1 Privacy

The MVP must not collect excessive personal information. Since there will be no need for permanent information storage, users must not be asked to give information that is not needed for demonstrating the seller-checking procedure (Bruton et al., 2023).

10.2 Consent

Users should know which information is being entered and why it is needed. The interface must not collect information about the seller and about the users without specific reasons.

10.3 Transparency

The user should be aware that the result of TrustCheck PK is based on information and signals. It must not be presented as a confirmation that a particular seller is totally safe or fraudulent.

10.4 Data Minimisation

The MVP requires information needed for the demonstration of the seller-checking procedure. Information storage, user profiles and any other personal information do not belong to the scope of the prototype.

10.5 User Control

TrustCheck PK is designed as a decision-support tool. The final purchase decision remains with the user. The application should provide understandable evidence and practical guidance rather than making an absolute decision on behalf of the consumer.

10.6 False Confidence and Risk

A low-risk result should not create a false sense of security. Even if the available evidence does not show major warning signs, users should still review seller information, payment methods, return policies and other relevant details before purchasing.

11. Final MVP User Journey

The final TrustCheck PK journey is designed to move the user from login through seller checking and evidence review to a final decision-support screen.

Login → Seller Check → Seller Details → Trust Snapshot → Evidence Breakdown →Safer Purchase Checklist → Decision Summary → Start Another Check

The journey serves the intended purpose of the MVP in providing support for the buyer to evaluate an unknown seller before making any payments. The journey is designed such that the user gets a chance to provide information about the seller, obtain an evaluation, review evidence, and obtain advice.

12. MVP Testing

The application was tested in the browser after development changes were made.

The main testing areas included:

- Login screen loads correctly.
- Seller information can be entered.
- Navigation between screens works correctly.
- Trust Snapshot displays correctly.
- Evidence information is visible.

- Safer Purchase Checklist is accessible.
- Decision Summary is displayed at the end.
- CustomButton works on multiple screens.
- CustomInput works on multiple screens.
- FormSection is reused across multiple screens.
- The 21st.dev component displays correctly.
- Existing functionality remains operational after component changes.

The application was also checked for consistent spacing, navigation, button behaviour and overall user flow.

13. Conclusion

TrustCheck PK was developed as an MVP to illustrate pre-purchase trust checking of social commerce users in Pakistan. The app offers a full user experience journey, from logging in, entering seller details to checking trustworthiness, reviewing evidence and guiding towards safer purchase decisions. The use of Cline was done in Plan Mode and Act Mode for inspection, planning and improvement of the application. This shows the necessary component-based development approach, as the application uses three reusable React components and one 21st.dev component. The MVP also takes into consideration privacy, consent, transparency, data minimization and possibility of developing a false sense of security.

References

Bruton, G. D, Pryor, C., & Cerecedo, J. A. (2023). Lean start-up in settings of impoverishment: The implications of the context for theory. *Journal of Management*, *50*(8). https://doi.org/10.1177/01492063231204869

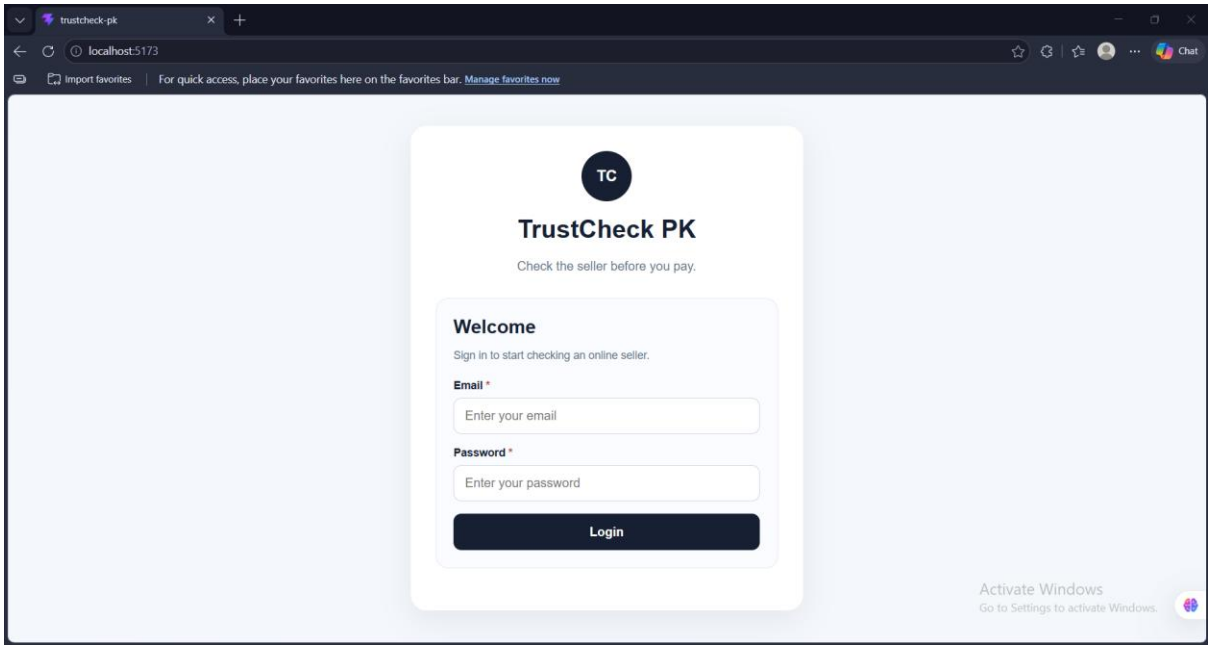
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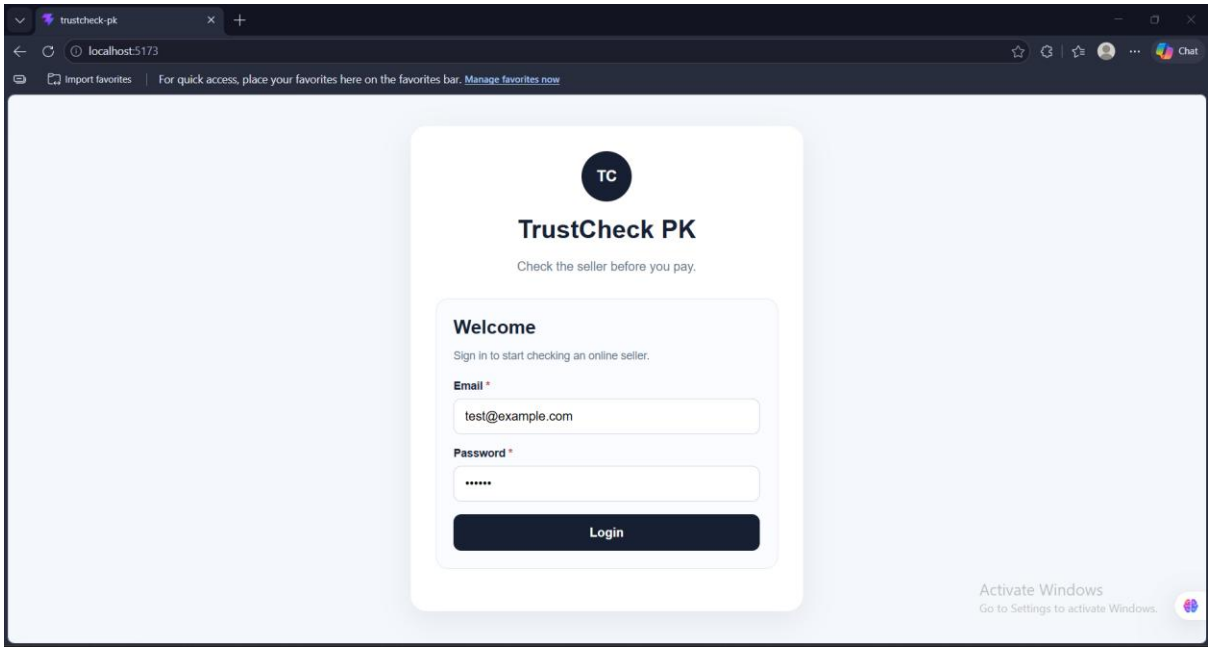
Lima, R. R., & da Silva, A. T. (2024). Accelerating the software development process through the application of MVP and monolithic architecture. *Proceedings of the XXIII Brazilian Symposium on Software Quality*, 469–477. https://doi.org/10.1145/3701625.3701632

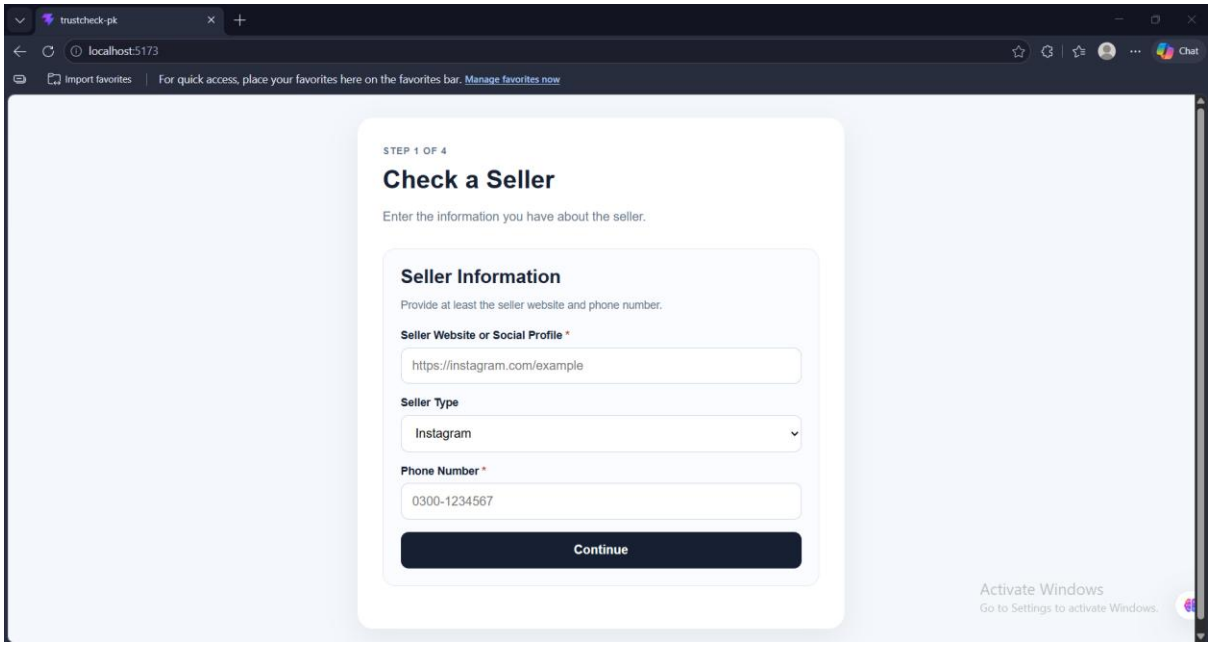
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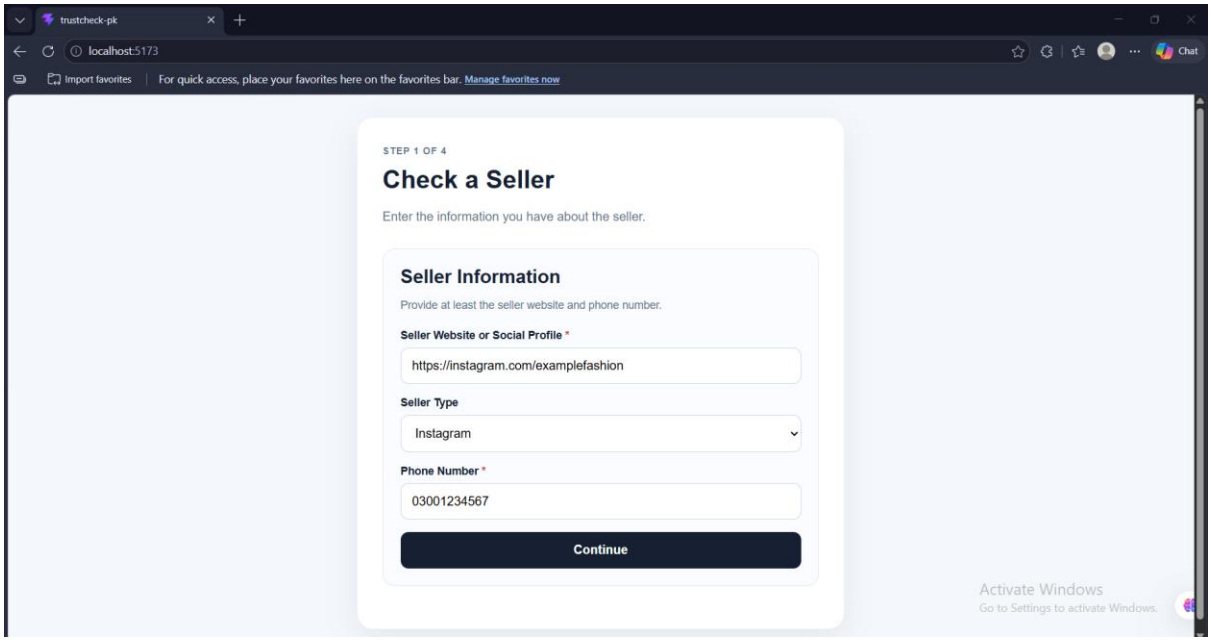
Appendix

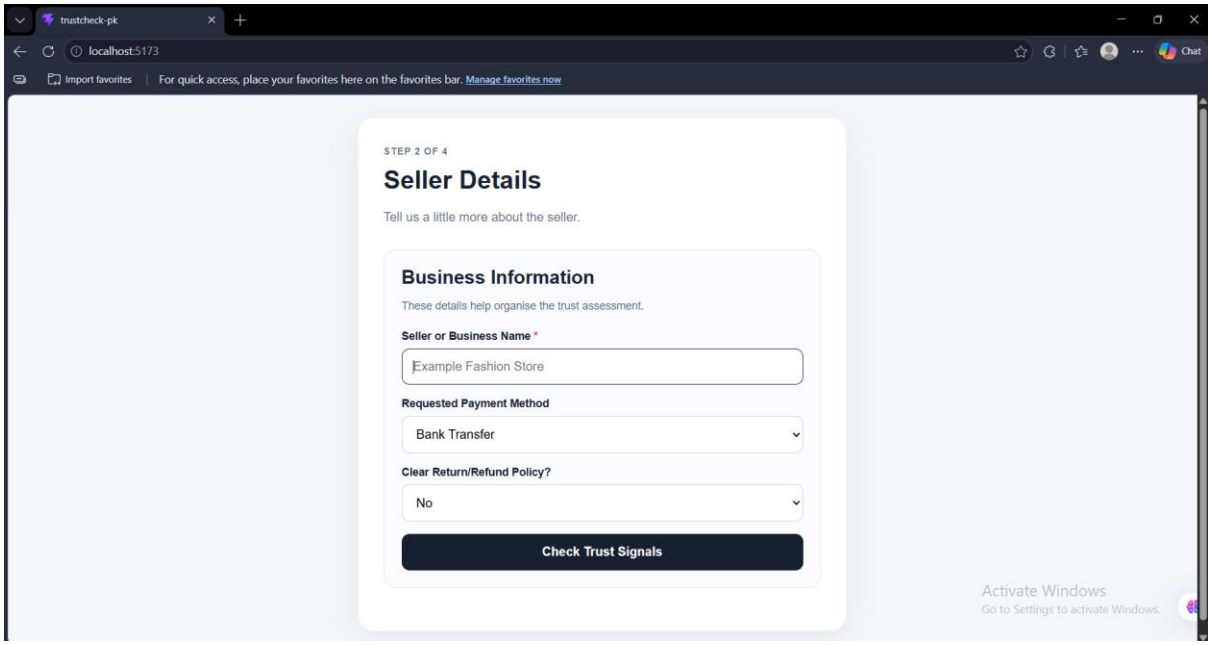
Test 1

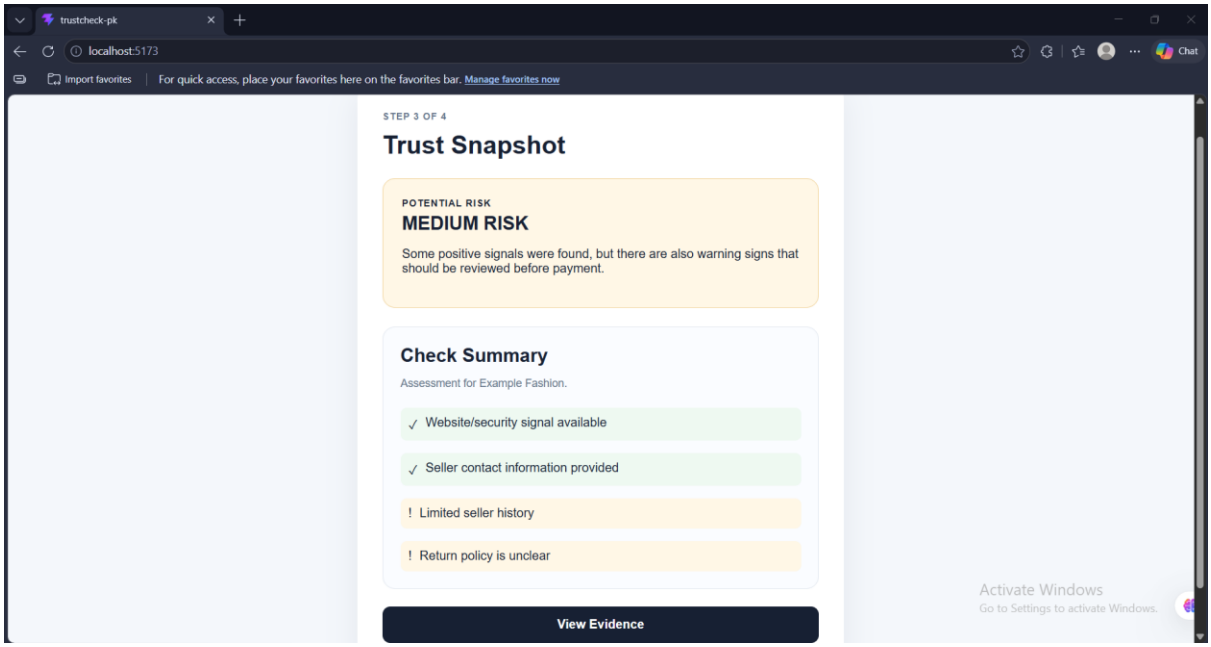

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
TC
TrustCheck PK
Check the seller before you pay.
Welcome
Sign in to start checking an online seller.
Email *
Enter your email
Password *
Enter your password
Login
Activate Windows
Go to Settings to activate Windows.

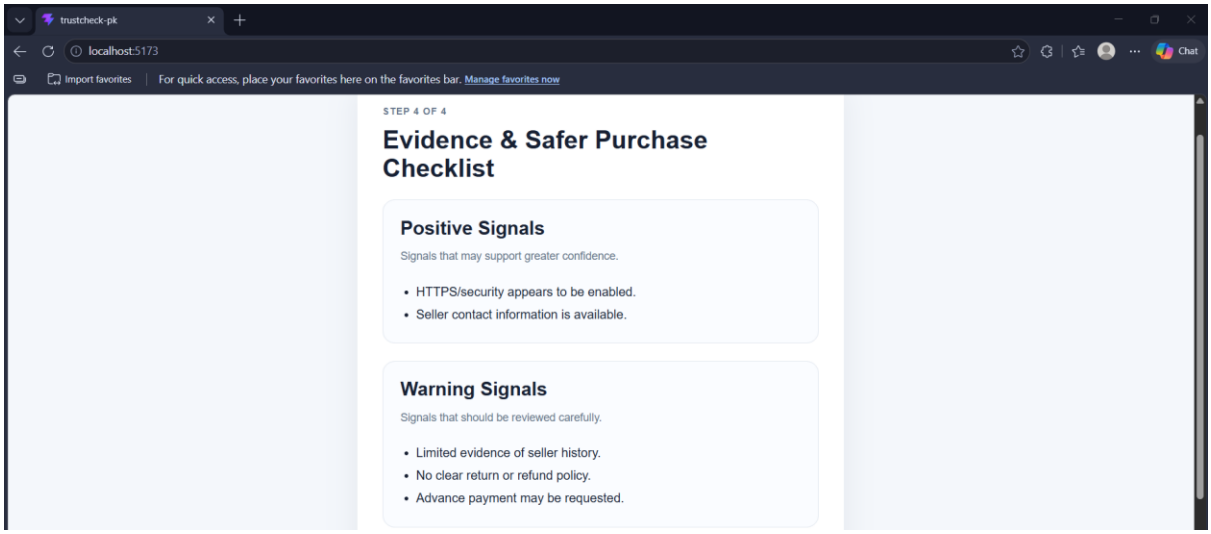

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
TC
TrustCheck PK
Check the seller before you pay.
Welcome
Sign in to start checking an online seller.
Email *
test@example.com
Password *
••••••
Login
Activate Windows
Go to Settings to activate Windows.

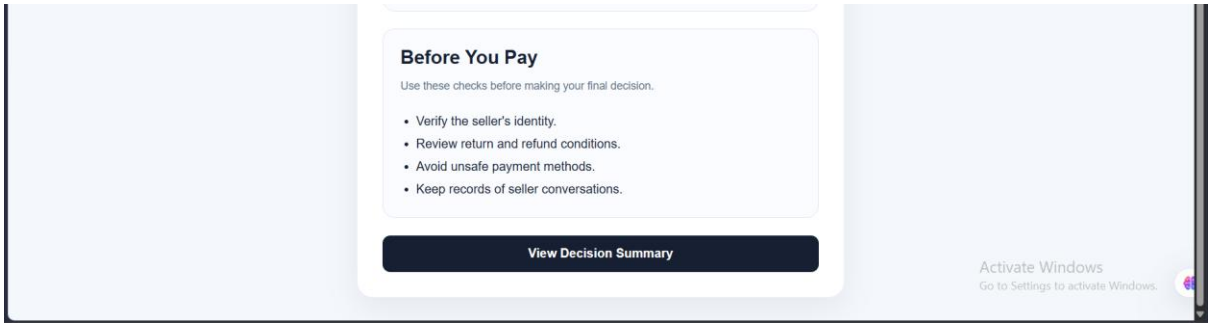

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
STEP 1 OF 4
Check a Seller
Enter the information you have about the seller.
Seller Information
Provide at least the seller website and phone number.
Seller Website or Social Profile *
https://instagram.com/example
Seller Type
Instagram
Phone Number *
0300-1234567
Continue
Activate Windows
Go to Settings to activate Windows.

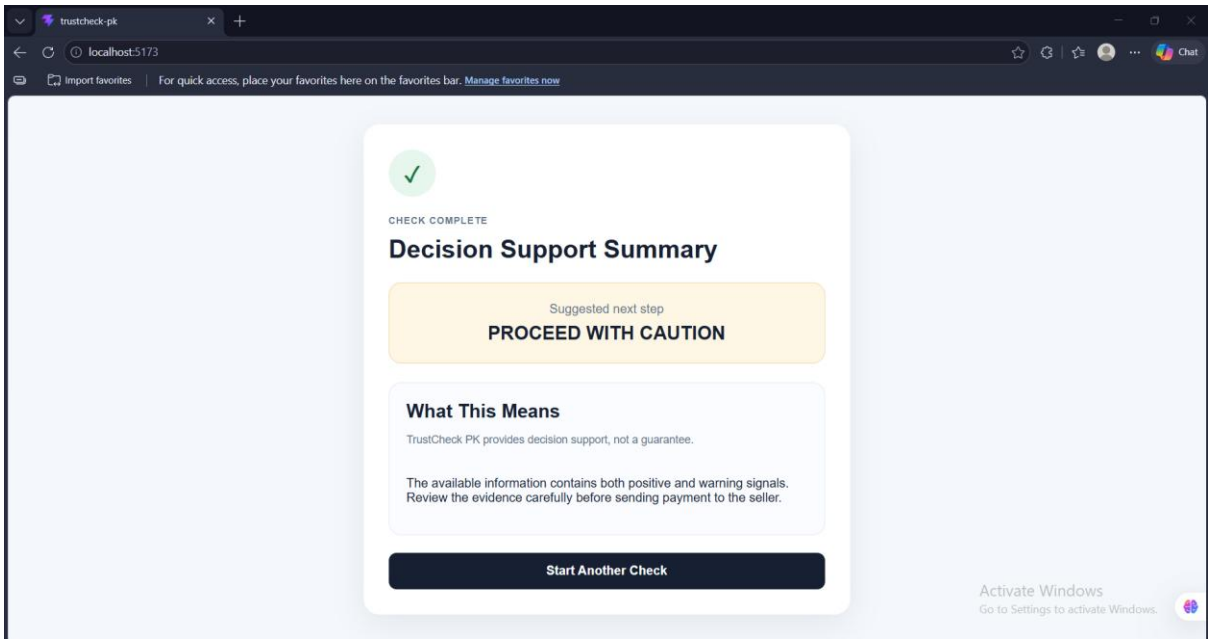

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
STEP 1 OF 4
Check a Seller
Enter the information you have about the seller.
Seller Information
Provide at least the seller website and phone number.
Seller Website or Social Profile *
https://instagram.com/examplefashion
Seller Type
Instagram
Phone Number *
03001234567
Continue
Activate Windows
Go to Settings to activate Windows.

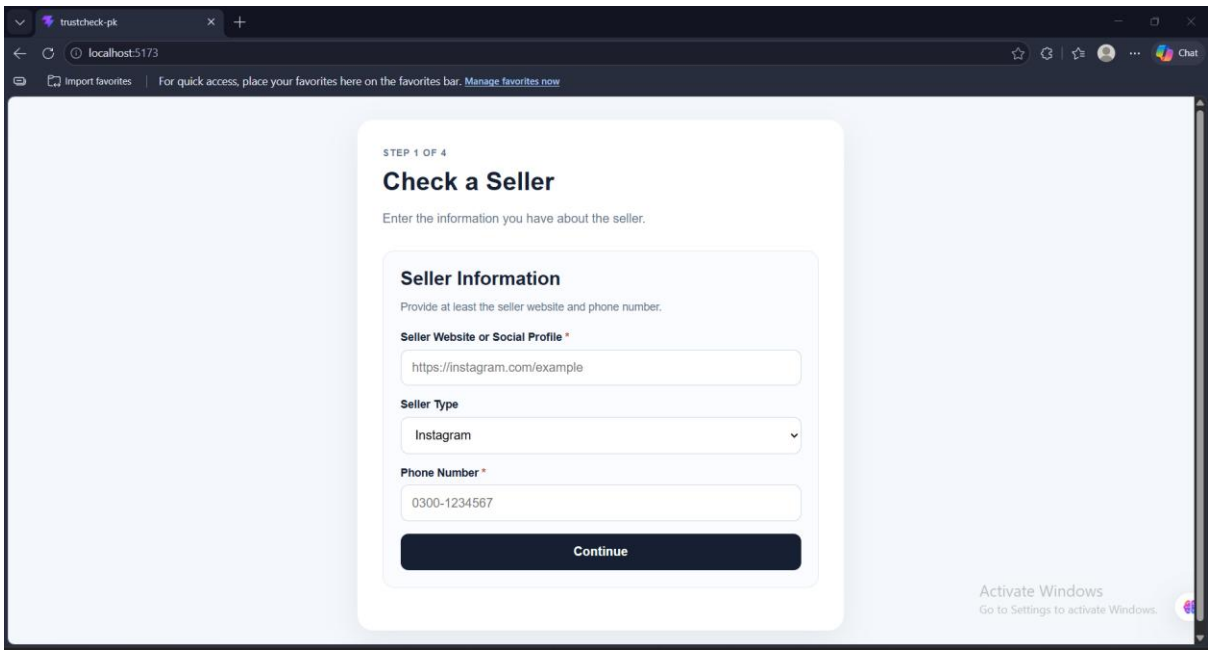

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
STEP 2 OF 4
Seller Details
Tell us a little more about the seller.
Business Information
These details help organise the trust assessment.
Seller or Business Name *
Example Fashion Store
Requested Payment Method
Bank Transfer
Clear Return/Refund Policy?
No
Check Trust Signals
Activate Windows
Go to Settings to activate Windows.


trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
STEP 3 OF 4
Trust Snapshot
POTENTIAL RISK
MEDIUM RISK
Some positive signals were found, but there are also warning signs that should be reviewed before payment.
Check Summary
Assessment for Example Fashion.
✓ Website/security signal available
✓ Seller contact information provided
! Limited seller history
! Return policy is unclear
View Evidence
Activate Windows
Go to Settings to activate Windows.

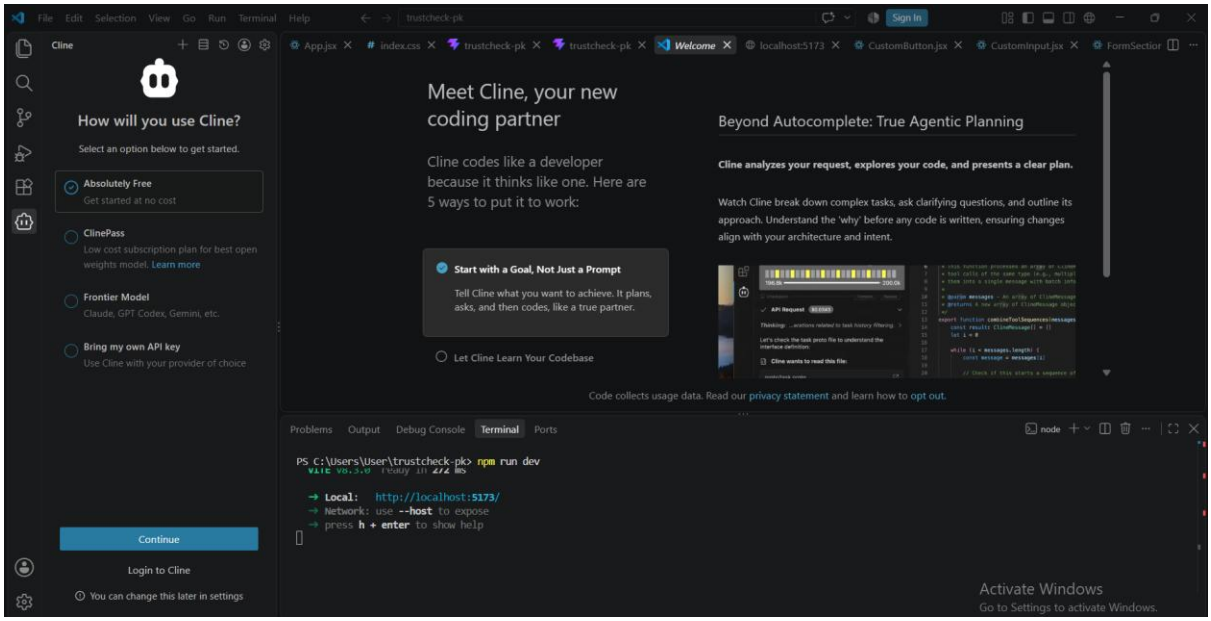

trustcheck-pk
localhost:5173
Import favorites
For quick access, place your favorites here on the favorites bar. Manage favorites now
Chat
STEP 4 OF 4
Evidence & Safer Purchase Checklist
Positive Signals
Signals that may support greater confidence.
• HTTPS/security appears to be enabled.
• Seller contact information is available.
Warning Signals
Signals that should be reviewed carefully.
• Limited evidence of seller history.
• No clear return or refund policy.
• Advance payment may be requested.



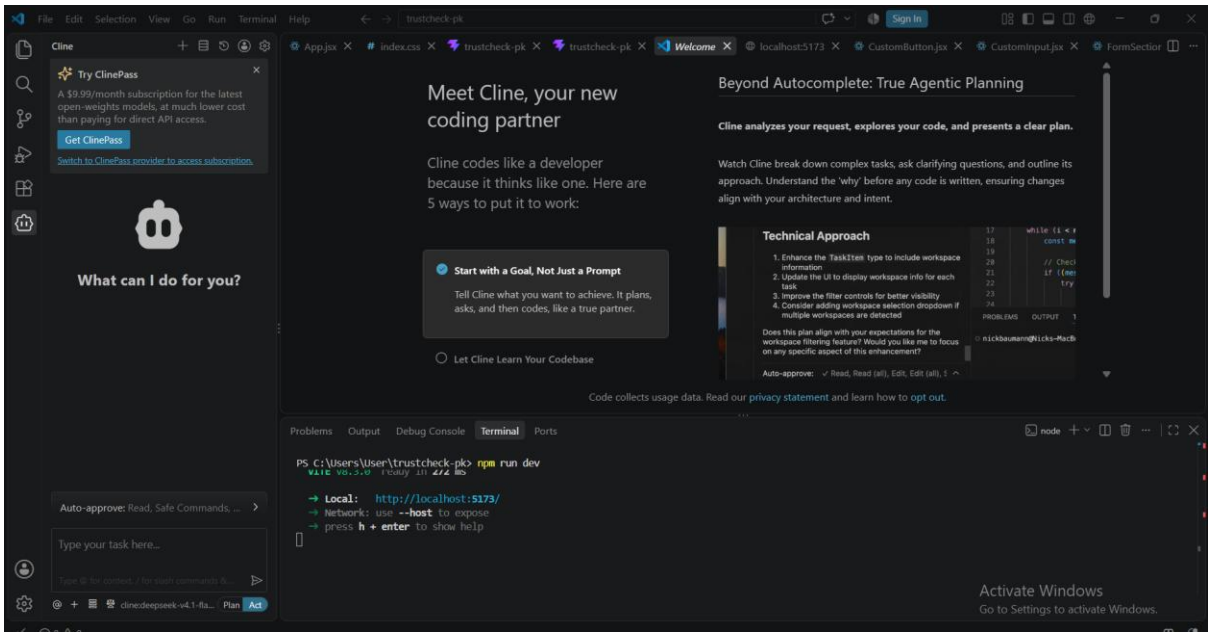




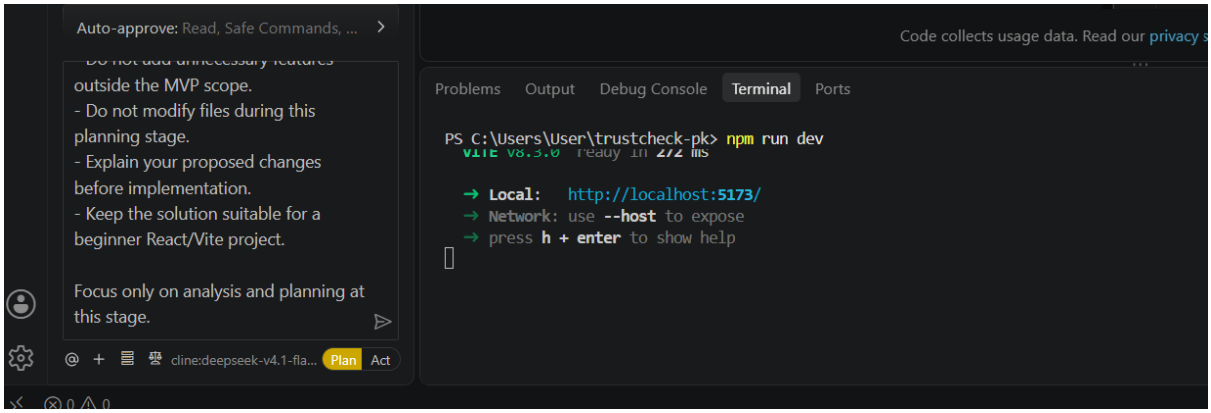
Setting up cline



Successfully setup



First prompt



Cline working

